고려대학교 약학대학
KOREA UNIVERSITY COLLEGE OF PHARMACY

KU-The Future
110 years and beyond

Microbiology

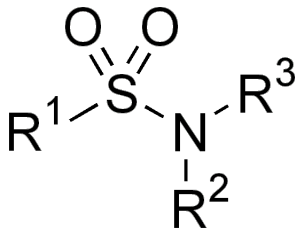


Fall semester, 2026

KU-The Future
110 years and beyond

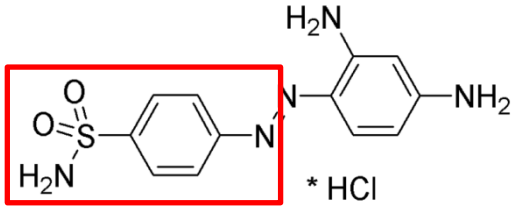
❖ Antimetabolites

▶ Sulfonamides (sulphonamide)

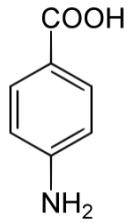


Sulfonamide
Functional group

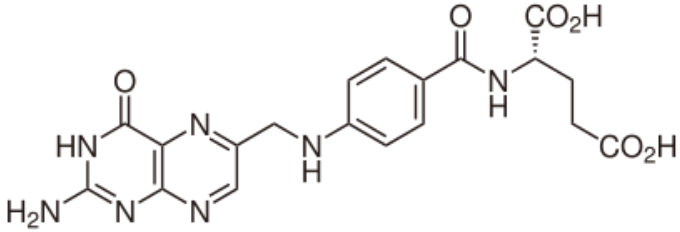
- synthetic antimicrobial agents containing the sulfonamide functional group
- <u>prontosil</u>:
  - a prodrug of sulfonamides / the first sulfonamide
  - the first commercially available antibiotic
  - be discovered by screening thousands of compounds related to azo dyes (by a research team at the Bayer Lab.)
- <u>mechanism of action</u>:
  - synthetic analogues of PABA → inhibiting dihydropteroate synthetase (PABA → dihydropteroic acid)
  - inhibiting the bacterial synthesis of folic acid



Prontosil (sulfonyl arylamine)

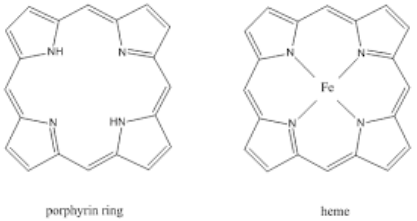


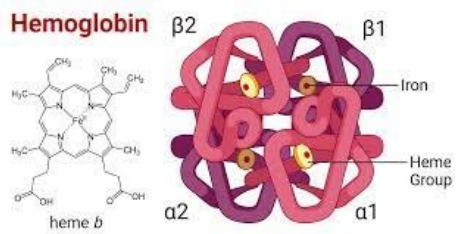
PABA



folic acid

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❖ Antimetabolites

▶ Sulfonamides (sulphonamide)

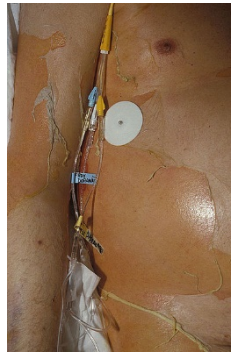
· spectrum of activity: most G(+) and many G(−)

· adverse effects:

- strong allergy reactions (hypersensitivity reactions)
  · Stevens-Johnson syndrome (SJS)
  · toxic epidermal necrolysis (TEN; 중독성표피박리)
  → flu-like symptoms / skin rash, blister / pneumonia, sepsis, multiorgan failure, etc.
- porphyria
- urinary tract disorders
- hemopoietic disorders



SJS

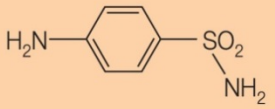


TEN

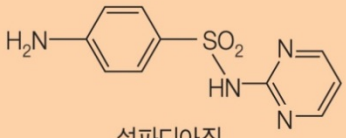
Antimetabolites

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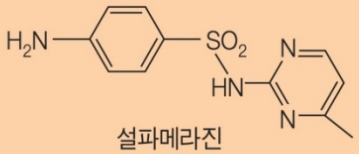
단기형 설파제



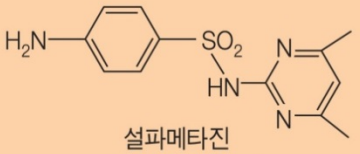
설파닐아마이드
(sulfanilamide)



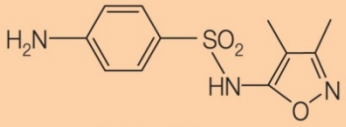
설파디아진
(sulfadiazine)



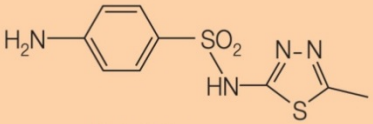
설파메라진
(sulfamerazine)



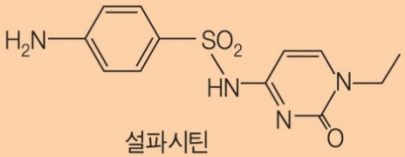
설파메타진
(sulfamethazine)



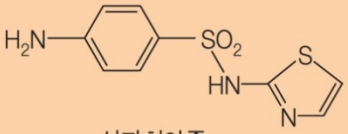
설피속사졸
(sulfisoxazole)



설파메치졸
(sulfamethizole)

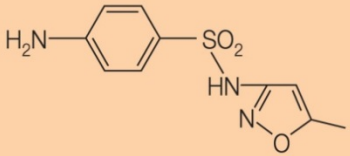


설파시틴
(sulfacytine)



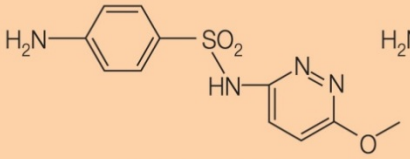
설파치아졸
(sulfathiazole)

중기형 설파제

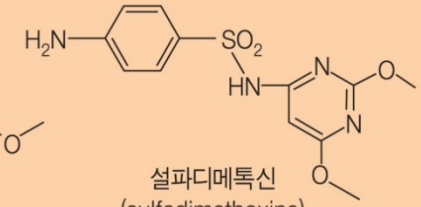


설파메톡사졸
(sulfamethoxazole)

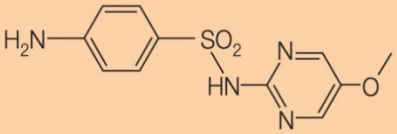
장기형 설파제



설파메톡시피리다진
(sulfamethoxypyridazine)



설파디메톡신
(sulfadimethoxine)



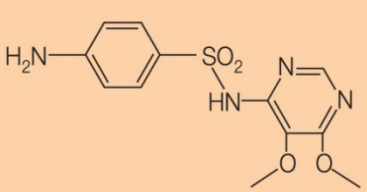
설파메톡시디아진
(sulfamethoxydiazine)

라이프사이언스
Life Science Publishing Co.

Antimetabolites

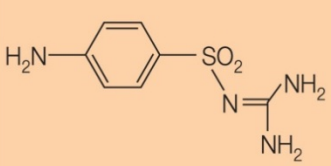
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110 years and beyond

초장기형 설파제

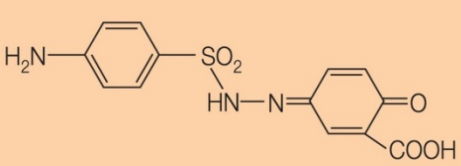


설파독신
(sulfadoxine)

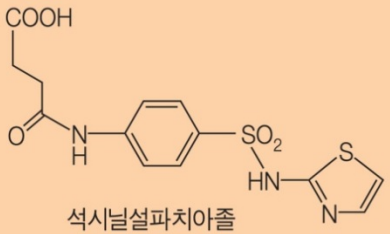
위장관에서 흡수되지 않는 설파제



설파구아니딘
(sulfaguanidine)

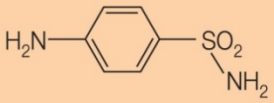


설파살라진
(sulfasalazine)

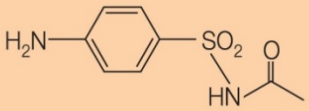


석시닐설파치아졸
(succinylsulfathiazole)

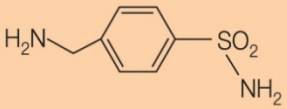
외용제로 사용되는 설파제



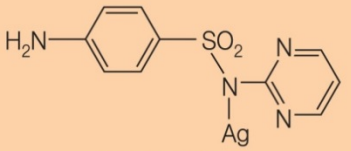
설파닐아마이드
(sulfanilamide)



설파아세트아미드
(sulfacetamide)



마페나이드
(mafenide)



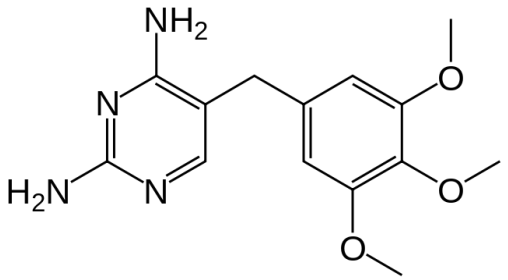
실버 설파디아진
(silver sulfadiazine)

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❖ Antimetabolites

▶ Trimethoprim (TMP)



· synthetic bacteriostatic antibiotic
  – be developed by George Hitchings

· <u>mechanism of action</u>:
  – inhibition of dihydrofolate reductase (dihydrofolic acid → tetrahydrofolic acid: essential precursor of purine and pyrimidine bases)

· <u>spectrum of activity</u>:
  – several species of G(–) and some G(+)
  – susceptible bacteria in orders Enterobacteriales (urinary *tract infections, Klebsiella pneumonia*, etc.) and Pasteurellales (*Haemophilus influenza*, etc.)

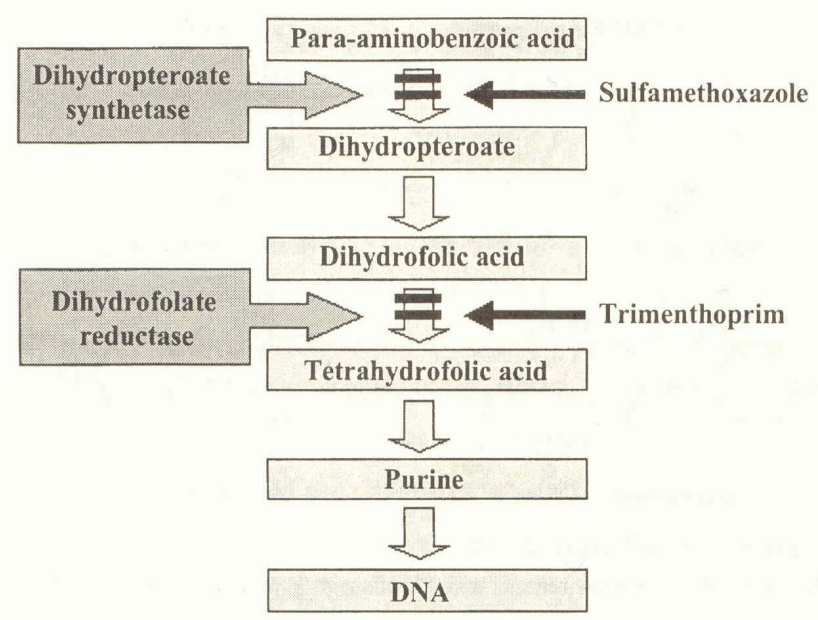
· <u>side effects</u>: nausea, vomiting, diarrhea / skin rash, itchiness, sun sensitivity / rarely, thrombopenia, etc.

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❖ Antimetabolites

▶ TMP-sulfa (TMP-SMX or co-trimoxazole)

- combination therapy of TMP and sulfa
  : both inhibit different steps in folate synthesis pathway
  → synergistic effect (greater efficacy; both G(+) and G(-))
- the combination's use: declining due to sulfonamide's side effects and resistance
- generally used for pneumonia, bronchitis; urinary tract infections; intestine infections, etc.
- folate: referring to many forms of folic acid and its related compounds, including (methyl)tetrahydrofolic acid, etc.

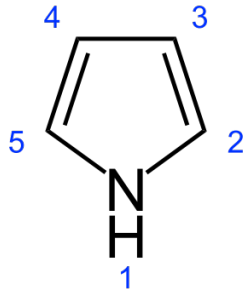


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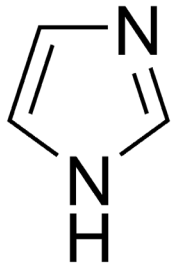
❖ Antifungal agents

▶ Aromatic heterocyles (5-membered ring)

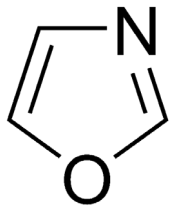
- members:
  - pyrrole: analogs with only one nitrogen atom in position 1
  - imidazole: having two nitrogen atoms
  - oxazole: analogs with the N in position 1 replaced by O
  - thiazole: analogs with the N in position 1 replaced by S
  - pyrazole: an analog with two adjacent Ns
  - triazole: analogs with three nitrogen atoms
- mechanism: interference with fungal membrane sterol biosynthesis
- be used topically in creams and solutions to control fungal skin infections



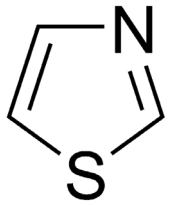
Pyrroles



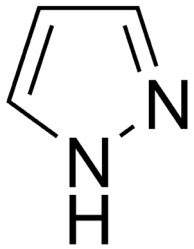
Imidazole



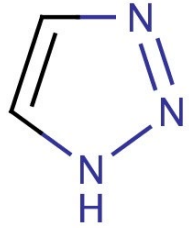
Oxazole



Thiazole



Pyrazole



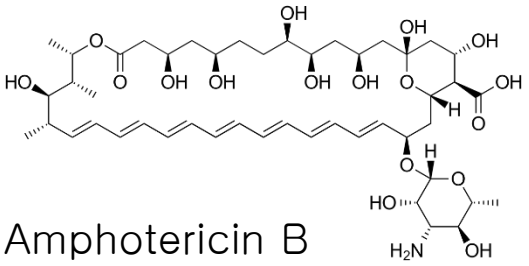
Triazole

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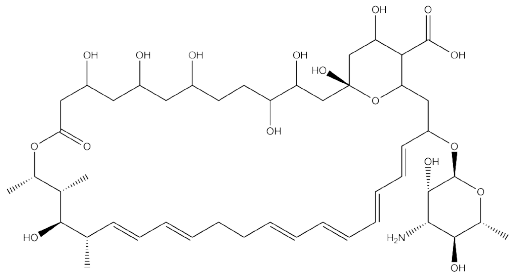
❖ Antifungal agents

▶ Polyenes

- poly-unsaturated organic compounds, containing at least three alternating double and single C-C bonds
- mechanism: binding to a particular fungal membrane sterol (ergosterol) → increase membrane permeability
- spectrum: be effective against fungi, and some algae and protozoa, but not plant and animal cells
- members:
  - **amphotericin B**: Fungizone / derived from *Streptomyces nodosus*
  - **nystatin**: Mycostatin / derived from *Streptomyces noursei*



Amphotericin B

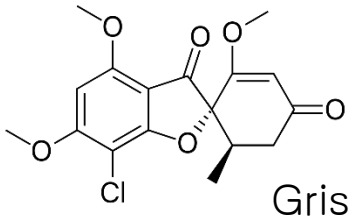


Nystatin

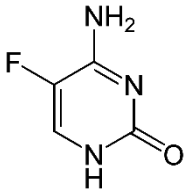
❖ Antifungal agents

▶ Others

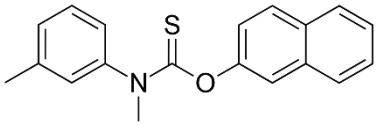
- **griseofulvin** (Fulvicin): from *Penicillium groseofulvum* / interfering with fungal mitosis / ringworm / not applying for people having porphyria or liver failure
- **flucytosine**: synthetic prodrug (flucytosine → 5-fluorouracil (by cytosine deaminase) / interfering with DNA & RNA synthesis / systemic fungal infections (e.g., candidiosis and cryptococcosis)
- **tolnaftate** (Tinactin): synthetic / unclear mode of action / ringworm
- **Terbinafine** (Lamisil): synthetic / interfering with sterol biosynthesis / ringworm



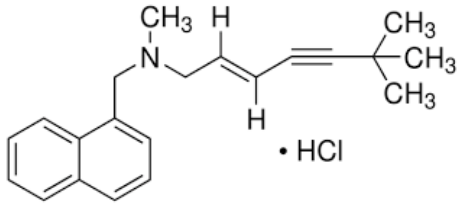
Griseofulvin



Flucytosine



Tolnaftate



Terbinafine